

Car Insurance Fraud Analysis

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Business Context

Sector : Property and Casualty (P&C) Insurance.

This project focuses on developing a Business Intelligence (BI) solution to analyze car insurance claims, detect fraud patterns, and understand cost drivers in the Property & Casualty insurance sector.

Project Goals

- Identify fraud patterns and high-risk segments
- Monitor KPIs such as Fraud Rate and Loss Ratio
- Optimize claims approval and settlement time
- Support data-driven decision-making

Business needs

- Undetected insurance fraud increases costs
- Inefficient manual investigation of low-risk claims
- Lack of centralized visibility over claims, drivers, and incidents

Key Research Questions

Understanding Fraud Rates and Influencing Factors

1. What is the overall fraud rate?
2. Does driver age impact fraud probability?
3. Are some vehicle categories more fraud-prone?
4. Do past offenders re-offend?
5. Does witness or police presence reduce fraud?

Process Summary

The project follows an end-to-end BI workflow

Using Python for data preparation and Power BI for dashboards.

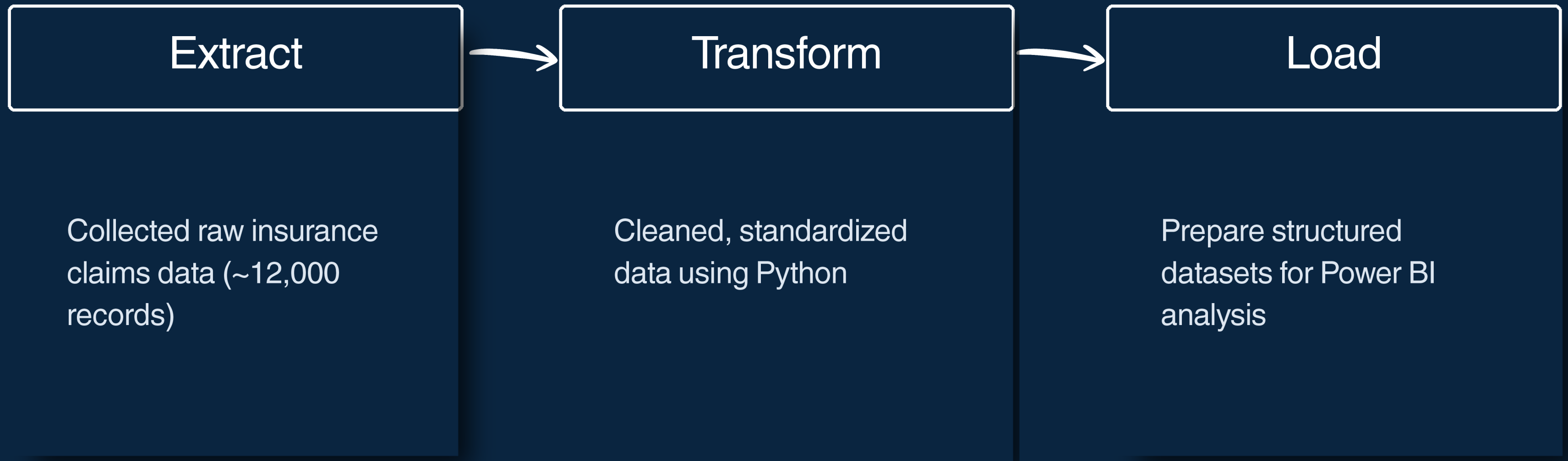
Data Gathering

ETL

Analysis

Visualization

ETL PROCESS



Data Preparation

Handling
missing values

```
# Replace '?' with NaN and impute
df.replace('?', np.nan, inplace=True)
df['age_of_driver'].fillna(df['age_of_driver'].median(), inplace=True)
df['gender'].fillna(df['gender'].mode()[0], inplace=True)
```

Standardizing
categorical fields

```
# Standardizing Gender text
df['gender'] = df['gender'].replace({'M': 'Male', 'F': 'Female'})

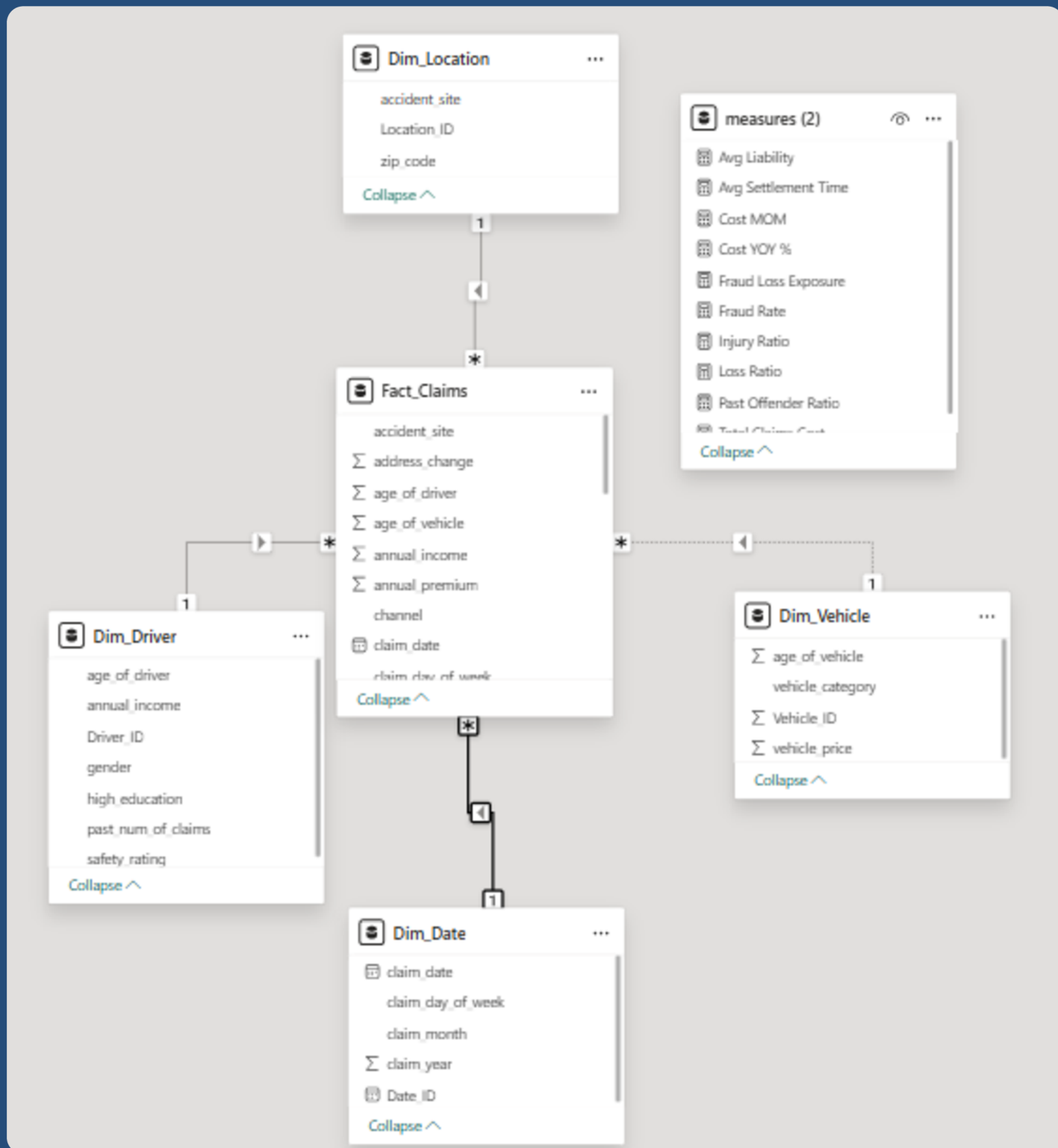
# Converting Fraud Reported to Binary (1/0)
df['fraud_reported'] = df['fraud_reported'].apply(lambda x: 1 if x == 'Y' else 0)
```

Time Series
Preparation

```
# Extracting Year and Month
df['claim_date'] = pd.to_datetime(df['claim_date'])
df['claim_year'] = df['claim_date'].dt.year
df['claim_month'] = df['claim_date'].dt.month_name()
```

Data Modeling & Storage Design

Star Schema design for analytical efficiency, optimized for Power BI performance



Identification of Fact and Dimension Tables

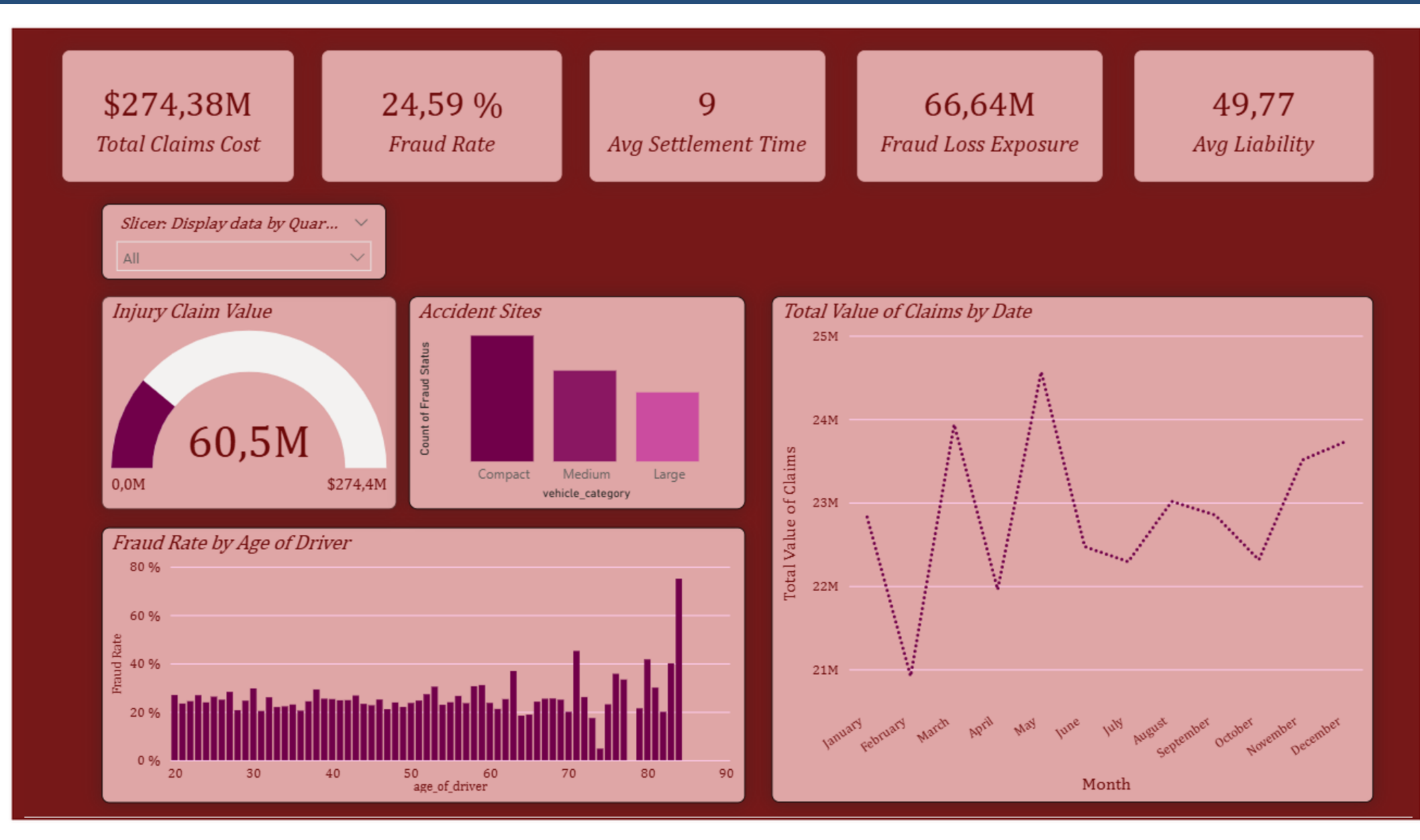
1. Fact Table:

The central table, it contains foreign keys and quantitative data (Measures) such as : claim amounts and counts

2. Dimension Tables:

- a. Location : Stores the geographical details of where the accidents occurred.
- b. Date : A lookup table used for time-intelligence analysis. It allows us to break down claim data by year, month, or day of the week.
- c. Driver: Contains demographic information about the policyholders.
- d. Vehicle: Provides specifics about the insured vehicles.

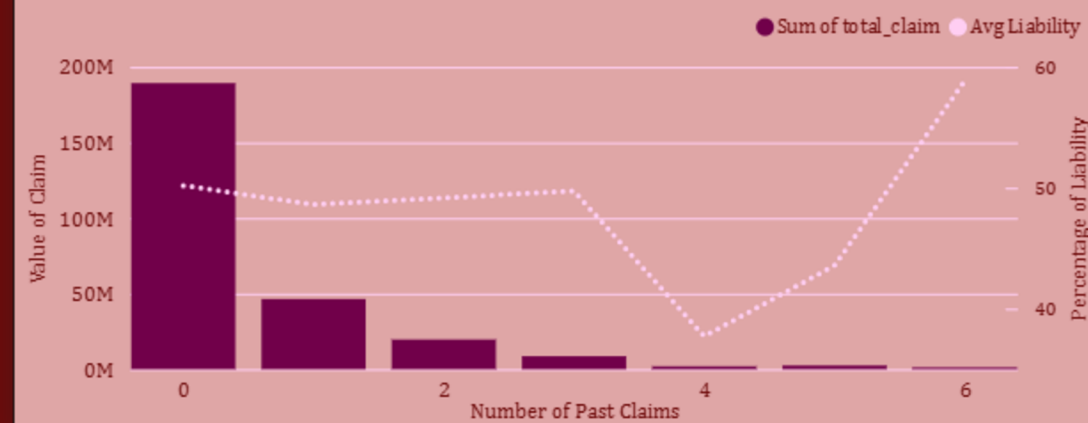
Data Visualization



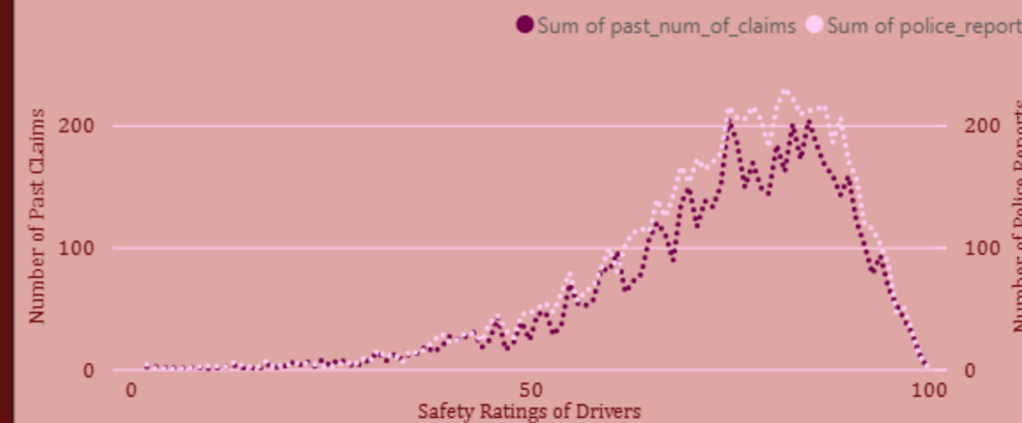
This dashboard provides a high-level view of the insurance portfolio's performance, highlighting key KPIs such as Total Claims Cost, Fraud Rate, Average Settlement Time, and Fraud Loss Exposure.

Executive Summary Dashboard

Value of Claim and Liability percentage by Number of Past Claims



Number of past claims by safety ratings of drivers



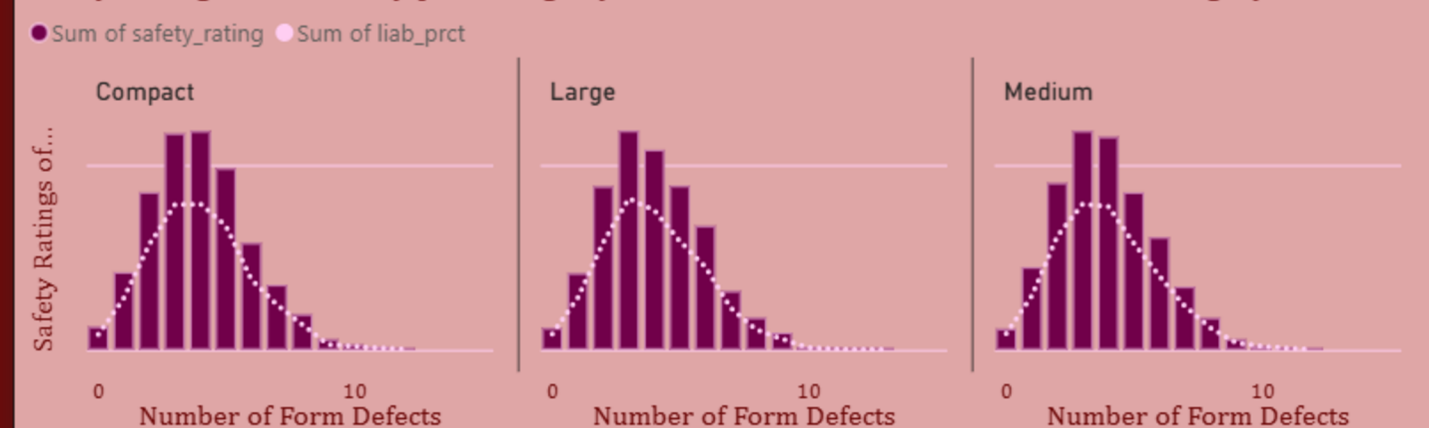
Vehicle Color

safety_rating

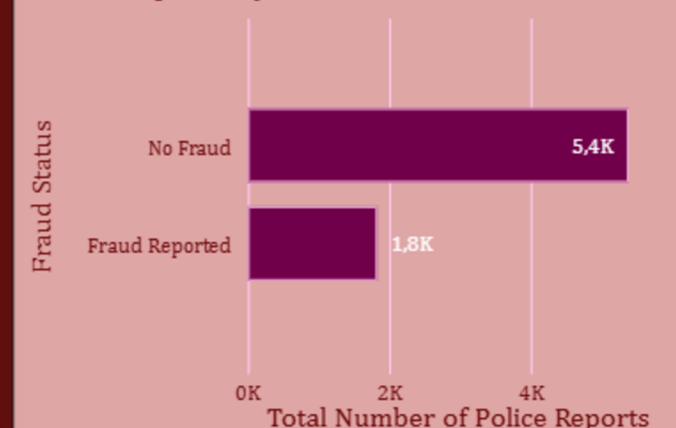
property_status

☐ Own
 ☐ Rent

Safety Ratings and Liability percentage by Number of Form Defects and Vehicle Category



Police Reports by Fraud Status



- Recidivism & Liability: Impact of claim history on costs
- Safety & Police Interaction: Link between safety ratings and police reports
- Vehicle Category: Safety and liability by vehicle class
- Fraud Verification: Police reports in fraud vs. legitimate claims

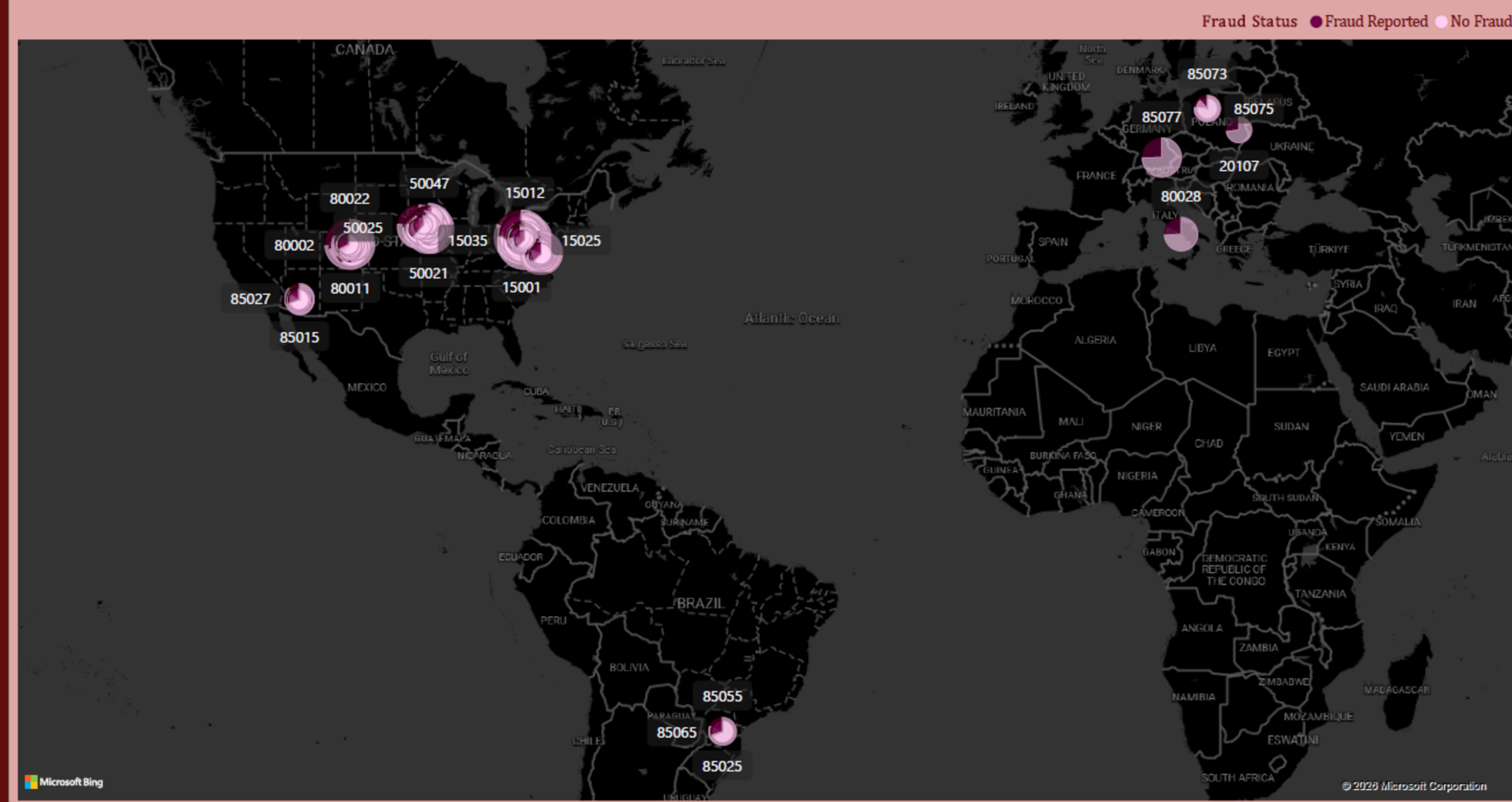
Deep Dive Analysis : General Overview



We isolated drivers who Own their vehicles and drive Red cars. This interaction instantly updates all metrics, revealing how specific demographics contribute to the overall risk profile.

Deep Dive Analysis : Filtered Scenario

Total Value of Claims by Location and Fraud Status



Using Azure Maps, this dashboard visualizes global and regional claim concentrations while using color-coded pie charts to instantly identify fraud hotspots across the US, Europe, South America, and East Asia.

Operational & Claim Details

Key Findings

High Financial Exposure: Total Fraud Loss Exposure of 66.64M, nearly a quarter of the portfolio's value is under threat.

Operational Lag: average settlement time stands at 9 days, indicating operational inefficiencies that contribute to higher costs.

Cost Drivers: Injury claims are a massive financial burden, totaling 60.5M, a significant portion of the 274.38M total claims cost.

Answers to Analytical Questions

Q1: Around 25% of claims flagged as fraudulent

Q2: Fraud risk fluctuates between middle-aged groups, spikes at 80+

Q3: Compact vehicles show higher fraud rates than large ones

Q4: Unwitnessed accidents are more likely fraudulent

Q5: Past offenders do tend to re-offend, the Average Liability Percentage tends upward with history

Business Decisions

Automated
Flagging

Implement a rule in the claims system to auto-flag any claim where Past Num of Claims > 0 for immediate review.

Premium
Adjustments

Adjust risk premiums for the "Compact" vehicle category to reflect the higher fraud exposure.

SIU Prioritization

Direct Special Investigation Unit (SIU) resources specifically toward unwitnessed accidents involving high injury claims.

Conclusion

This BI project demonstrates how data-driven insights can reduce fraud exposure, improve operational efficiency, and support strategic decision-making in the insurance sector.

Thank You